

Part 0 : Preparation

Starting by entering the deployment space to the overview page

The screenshot shows the IBM Watsonx Overview page for the deployment space 'lab 4 final 3'. The page has a dark header with the IBM Watsonx logo and user information. Below the header, there's a navigation bar with tabs: Overview (selected), Assets, Deployments, Jobs, and Manage. The main content area is divided into three sections: 'Jump back in' on the left, 'Deployments' in the center, and 'Space history' on the right. The 'Jump back in' section lists recent assets like 'final-P2_hpo_d_output' and 'dataset_loan_example_with_loanID_Ylabel4 copy.csv'. The 'Deployments' section shows a summary of 1 deployed and 0 failed deployments, and 0 active and 0 failed job runs. The 'Space history' section shows 'No notifications'.

Then, go to the deployments tab and click on the deployment which you have deployed earlier

The screenshot shows the IBM Watsonx Deployments tab for the deployment space 'lab 4 final 3'. The page has a dark header with the IBM Watsonx logo and user information. Below the header, there's a navigation bar with tabs: Deployments (selected) and AI Factsheet. The main content area is a table listing deployments. The table has columns: Name, Type, Status, Tags, and Last modified. There is one deployment listed: 'Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8' with status 'Online' and 'Deployed'. The right sidebar shows details for the selected deployment, including 'About this asset', 'Name', 'Description', 'Asset Details', and 'Tags'.

Name	Type	Status	Tags	Last modified
(ip) Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8	Online	Deployed		1 minute ago Chananop Wannatipaporn (You)

Once you clicked on the deployment, you should be entering this page

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8 Deployed Online

API reference Test Evaluations Transactions AI Factsheet

Endpoints for inferencing ⓘ

Private endpoint

`https://private.us-south.ml.cloud.ibm.com/ml/v4/deployments/d1b9903c-9284-4c3d-ae19-37e8b9f63bfa/predictions?version=2021-05-01`

Public endpoint

`https://us-south.ml.cloud.ibm.com/ml/v4/deployments/d1b9903c-9284-4c3d-ae19-37e8b9f63bfa/predictions?version=2021-05-01`

Learn more about the 2021-05-01 version query parameter

Code snippets

cURL	Java	JavaScript	Python	Scala
<pre># NOTE: you must set \$API_KEY below using information retrieved from your IBM Cloud account (https://dataplatform.cloud.ibm.com/docs/content/wsj/analyze-data/ml-authenticating.html) curl -X POST -H "Content-Type: application/x-www-form-urlencoded" -H "Accept: \ application/json" -d "grant_type=urn:ibm:params:oauth:grant-type:apikey" \ --data-urlencode "apikey=\$API_KEY" "https://iam.cloud.ibm.com/identity/token" # the above CURL request will return an auth token that you will use as \$IAM_TOKEN in the scoring request below # TODO: manually define and pass values to be scored below curl -X POST -H "Content-Type: application/json" -H "Accept: application/json" -H "Authorization: \ Bearer \$IAM_TOKEN" -d '{"input_data": [{"fields": [{"ARRAY_OF_INPUT_FIELDS}], "values": [{"ARRAY_OF_VALUES_TO_BE_SCORED}], \ [ANOTHER_ARRAY_OF_VALUES_TO_BE_SCORED]]}]' "https://private.us-south.ml.cloud.ibm.com/ml/v4/deployments/d1b9903c-9284-4c3d-ae19-37e8b9f63bfa/predictions?version=2021-05-01"</pre>				

About this deployment

Name

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Description

No description provided.

Deployment Details

Deployment ID: d1b9903c-9284-4c...

Serving name:

No serving name.

Software specification:

hybrid_0.1

Hybrid pipeline software specifications:

autoai-kb_r24.1-py3.11

Copies:

1

Tags

Add tags to make assets easier to find.

Associated asset

final-P2_hpo_d_output

822a356c-baea-4fcd-a098-12d32ba84ef8

Last modified

1 minute ago

Created on

Oct 7, 2024

Part 1 : Configuring model evaluations

Click on Evaluations tab then click on Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8 Deployed Online

API reference Test **Evaluations** Transactions AI Factsheet

Evaluation results

View Watson OpenScale evaluation results to gain insights about your model performance

Start configuring model evaluations

To view evaluation results, configure evaluations to monitor the deployment for fairness or performance drift.

[Configure OpenScale evaluation settings](#)

Once you click on the Configure OpenScale evaluation settings, you should be entering this page. Set the data type and algorithm type according to your data asset. When done click the View summary button.

IBM watsonx

Deployments / for lab 4 v2 / with-P2_hpo_d_output /

Prepare for evaluation

Provide deployment details to determine the appropriate configuration settings.

Provide model information

[Show less](#)

Details
Specify the data type and algorithm type for the model and provide the endpoint location if available.

☐ Import settings

Data type
Numeric/categorical

Algorithm type
Binary classification

Endpoint location
<https://private.us-south.ml.cloud.ibm.com/ml/v4/deployments/09f8b4c3-8648-4695-85b3-b71cd2da23dd/prec>

[Cancel](#) [Back](#) [View summary](#)

After clicking view summary, check your deployed model, model input data type, algorithm type and feature columns whether it's correct or not. Once it's all correct click save and continue.

IBM watsonx

Deployments / lab 4 final 3 / final-P2_hpo_d_output /

Review setup for Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Review the model summary and click Save and continue to proceed.

Deployed model	Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8
Model input data type	Numeric/categorical
Algorithm type	Binary classification
Feature columns	ApplicantIncome, CoapplicantIncome, Credit_History, Dependents, Education, Gender, Loan_Amount_Term, LoanAmount, LoanAmount_Term, Married, Property_Area, Self_Employed

[Cancel](#) [Back](#) [Save and continue](#)

Once you click save and continue, you should now be at this page. Select your configuration method as “Use manual setup” then click next.

Configure OpenScale evaluation settings

Configure Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Select a configuration method to prepare for model evaluation.

Select configuration method

Details

There are two setup options:

Use manual setup

Configure your model evaluations manually by completing steps that prepare your model for evaluation.

Upload configuration package

Run a notebook to generate a file that you can upload to configure evaluations.

Configuration method

☒ Use manual setup

☐ Upload configuration package

Cancel Back Next

After that, upload your training data. However, the label column of the data file must not include a unique value and need to be named as ‘y’ as in the example data format below

Configure OpenScale evaluation settings

Configure Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Select a configuration method to prepare for model evaluation.

Specify training data

Details

Upload or connect to training data that is stored in a database or cloud storage to configure model evaluations.

Training data option

Upload file

Provide a CSV file containing the training data.

Drag training data here or [browse](#) (8 MB size limit)

Cancel Back Next

Data format should have no unique value and required the label column to be named as 'Y'

dataset_loan_example_v2

Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Y
Male	No	0	Graduate	No	5849	0		360	1	Urban	Y
Male	Yes	1	Graduate	No	4583	1508	128	360	1	Rural	N
Male	Yes	0	Graduate	Yes	3000	0	66	360	1	Urban	Y
Male	Yes	0	Not Graduate	No	2583	2358	120	360	1	Urban	Y
Male	No	0	Graduate	No	6000	0	141	360	1	Urban	Y
Male	Yes	2	Graduate	Yes	5417	4196	267	360	1	Urban	Y
Male	Yes	0	Not Graduate	No	2333	1516	95	360	1	Urban	Y
Male	Yes	3	Graduate	No	3036	2504	158	360	0	Semiurban	N
Male	Yes	2	Graduate	No	4006	1526	168	360	1	Urban	Y
Male	Yes	1	Graduate	No	12841	10968	349	360	1	Semiurban	N
Male	Yes	2	Graduate	No	3200	700	70	360	1	Urban	Y
Male	Yes	2	Graduate		2500	1840	109	360	1	Urban	Y
Male	Yes	2	Graduate	No	3073	8106	200	360	1	Urban	Y
Male	No	0	Graduate	No	1853	2840	114	360	1	Rural	N
Male	Yes	2	Graduate	No	1299	1086	17	120	1	Urban	Y
Male	No	0	Graduate	No	4950	0	125	360	1	Urban	Y
Male	No	1	Not Graduate	No	3596	0	100	240		Urban	Y
Female	No	0	Graduate	No	3510	0	76	360	0	Urban	N
Male	Yes	0	Not Graduate	No	4887	0	133	360	1	Rural	N
Male	Yes	0	Graduate		2600	3500	115		1	Urban	Y
Male	Yes	0	Not Graduate	No	7660	0	104	360	0	Urban	N

After you are done uploading the data file, select the delimiter according to your file format. Which for .csv is Comma (.). Then click next to continue onto the next page.

IBM watsonx

Deployments / for lab 4 v2 / with-P2_hpo_d_output /

Configure OpenScale evaluation settings

Configure with loan id y label

Select a configuration method to prepare for model evaluation.

Model View

Model Specify training data

Endp Select features and label

Expl Select model output

Gen

SHA1

LIME

Evalu

Fairm

Qual

Drift

Drift

Gen

Model

Cust

my n

Specify training data

Details

Upload or connect to training data that is stored in a database or cloud storage to configure model evaluations.

Training data option

Upload file

Training data file

dataset_loan_example_with_loanID_Ylabel.csv

Select delimiter

Comma (,)

Cancel

Back

Next

Once you are done with the previous step, now you should be landing on this page. Select the column type for feature column (categorical/non-categorical)

Configure Deployment for model 94b076c2-b087-4769-9b2c-b2016ff56cb3

Select a configuration method to prepare for model evaluation.

Select the feature columns and label column

Details

From the training data, select the input features. Indicate if numeric feature values represent categories by checking the Categorical checkbox. Next, select the label column. The label represents the correct prediction (ground-truth) for each record. A column can be marked as a feature or label but not both.

Select the feature columns

Selected features 12

Features (13)	Type	Categorical	Label / Target
☒ ApplicantIncome	000	☐	☐
☒ CoapplicantIncome	000	☐	☐
☒ Credit_History	000	☒	☐
☒ Dependents	000	☐	☐
☒ Education	A	☒	☐
☒ Gender	A	☒	☐

Items per page: 25 1 - 13 of 13 items

Cancel Back Next

Then select label column which in this place is 'Y' by ticking in this area

Configure Deployment for model e7c394e4-11b5-4048-9bab-d3983bc5c7a5

Select a configuration method to prepare for model evaluation.

Select the feature columns and label column

Details

From the training data, select the input features. Indicate if numeric feature values represent categories by checking the Categorical checkbox. Next, select the label column. The label represents the correct prediction (ground-truth) for each record. A column can be marked as a feature or label but not both.

Select the feature columns

Selected features 11

☒ Self_Employed	A	☒	☐
☒ ApplicantIncome	000	☐	☐
☒ CoapplicantIncome	000	☐	☐
☒ LoanAmount	000	☐	☐
☒ Loan_Amount_Term	000	☐	☐
☒ Credit_History	000	☐	☐
☒ Property_Area	A	☒	☐
☐ Y	A	☐	☒

Items per page: 25 1 - 12 of 12 items

Cancel Back Next

Once you are done marking all the features and label columns to its type, click next.

Configure Deployment for model e7c394e4-11b5-4048-9bab-d3983bc5c7a5

Select a configuration method to prepare for model evaluation.

Select the feature columns and label column

Details

From the training data, select the input features. Indicate if numeric feature values represent categories by checking the Categorical checkbox. Next, select the label column. The label represents the correct prediction (ground-truth) for each record. A column can be marked as a feature or label but not both.

Select the feature columns

Feature	Type	Categorical	Selected
☒ Self_Employed	📄	☒	☐
☒ ApplicantIncome	🔢	☐	☐
☒ CoapplicantIncome	🔢	☐	☐
☒ LoanAmount	🔢	☐	☐
☒ Loan_Amount_Term	🔢	☐	☐
☒ Credit_History	🔢	☐	☐
☒ Property_Area	📄	☒	☐
☐ Y	📄	☐	☒

Items per page: 25 1 - 12 of 12 items

Next

Then continue with selecting the model output by marking prediction as 'Prediction' and probability as 'Probability'.

Configure Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Select a configuration method to prepare for model evaluation.

Select model output

Details

From the model output data, select the column that contains the prediction generated by the deployed model. Select the prediction probability column which contains the model's confidence in the prediction.

Select the prediction and probability column(s).

Features (2)	Type	Prediction	Probability
prediction	📄	☒	☐
probability	🔢	☐	☒

No probability column selected
To support explainability, drift detection, and debiasing for binary classification you must select one or two prediction probability columns

Back View summary

Once done marking, click 'View Summary'.

Configure OpenScale evaluation settings

Configure Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Select a configuration method to prepare for model evaluation.

Select configuration method

- Select configuration method
- Specify training data
- Select features and label
- Select model output

Select model output

Details

From the model output data, select the column that contains the prediction generated by the deployed model. Select the prediction probability column which contains the model's confidence in the prediction.

Select the prediction and probability column(s).

Filter table

Features (2)	Type	Prediction	Probability
prediction	📄	☒	
probability	📄		☒

Cancel Back View summary

Check all your setup and make sure that what you choose aligns with your dataset. Once done, click finish.

Configure OpenScale evaluation settings

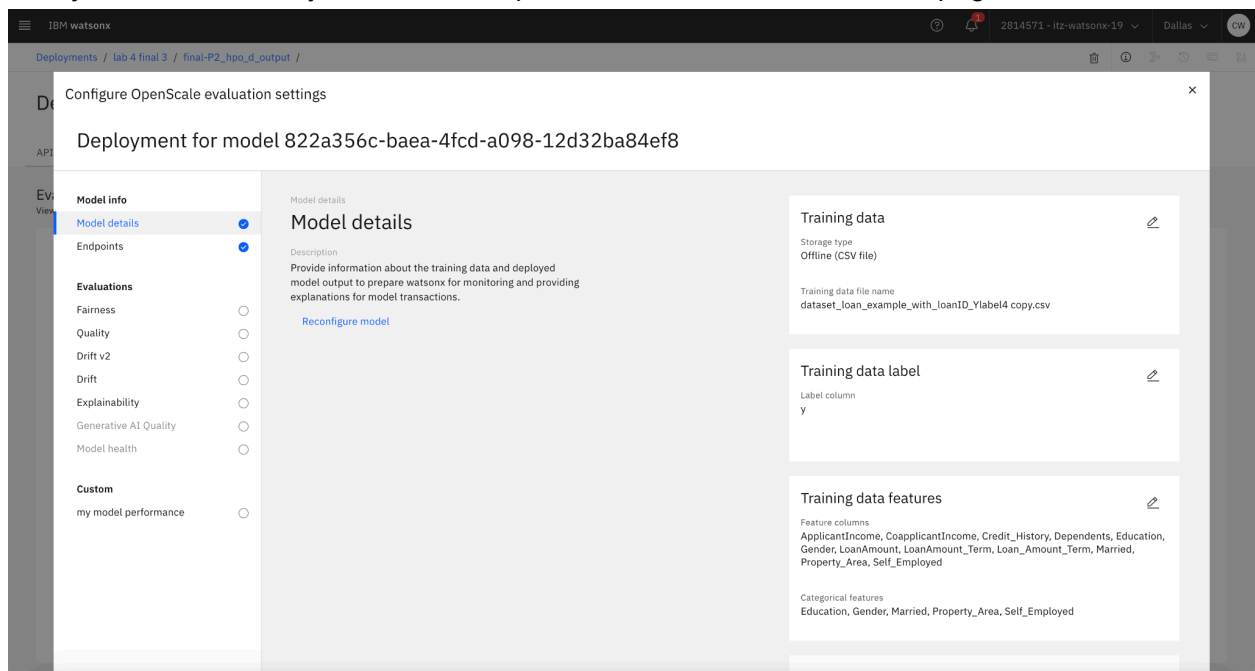
Complete setup for Deployment for model e7c394e4-11b5-4048-9bab-d3983bc5c7a5

Review the model summary and click Finish to complete setup.

Deployed model	Deployment for model e7c394e4-11b5-4048-9bab-d3983bc5c7a5
Model input data type	Numeric/categorical
Algorithm type	Binary classification
Training data	dataset_loan_example_v2.csv
Label column	Y
Feature columns	ApplicantIncome, CoapplicantIncome, Credit_History, Dependents, Education, Gender, Loan_Amount_Term, LoanAmount, Married, Property_Area, Self_Employed
Categorical features	Education, Gender, Married, Property_Area, Self_Employed
Prediction column	prediction
Probability column	probability

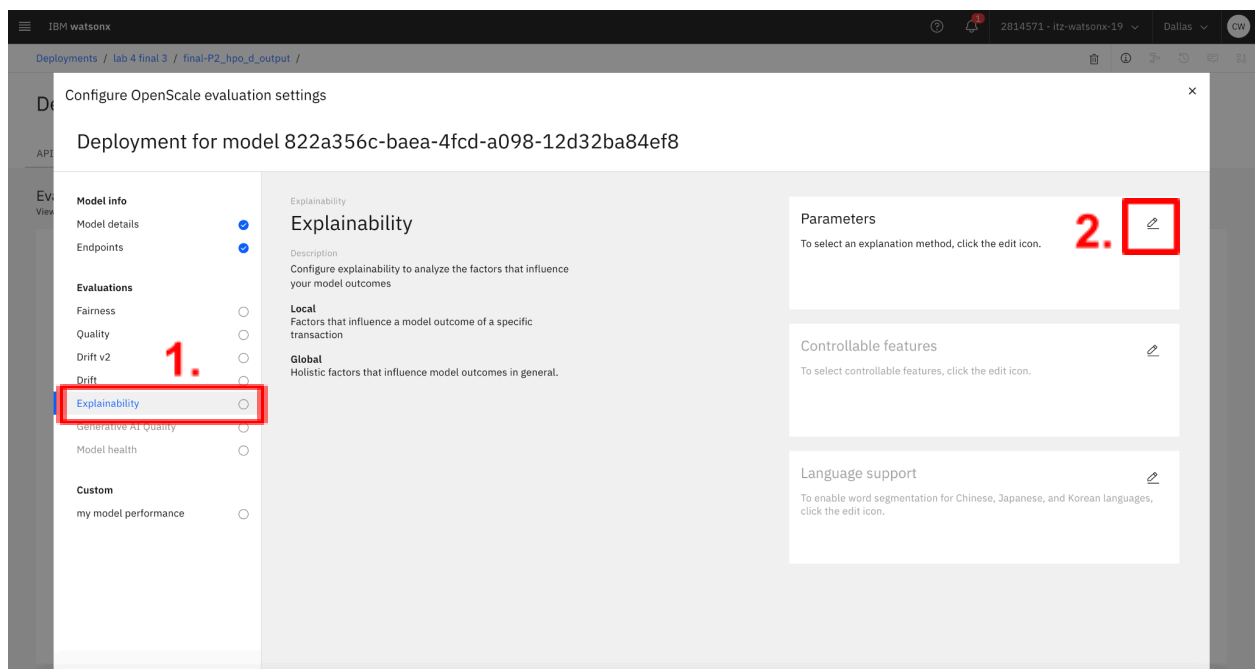
Cancel Back Finish

Now you have finished your model setup and should be able to see this page.



Part 2 : Configuring model explainability

Go to general settings under the explainability topic and you should be able to see this page. After that click on the edit icon of the explanation method to start configuring the model explanation method.



Then choose 'LIME (enhanced)' which is a local explanation as an explanation method and click next to continue onto the next page.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
- Endpoints

Evaluations

- Fairness
- Quality
- Drift v2
- Drift
- Explainability**
- Generative AI Quality
- Model health

Custom

- my model performance

Explainability Parameters

Description

OpenScale uses SHAP or LIME (enhanced) to calculate the feature influence and reveals which features contributed to the model's predicted outcome.

SHAP assigns each model feature an importance value called a SHAP value. The SHAP values are the estimate of the average marginal contribution of a feature value across all possible groups of features. The sum of input features SHAP values is the difference between average model prediction and the prediction of the transaction being explained. SHAP requires access to the model scoring endpoint and scores the perturbations for each local explanation.

LIME (enhanced) generates explanations by modifying model input and observing the model's response around the vicinity of the transaction and creating a local linear model using the perturbations. LIME (enhanced) explanations for structured data could be generated using pre scored perturbations, which allows scoring the model once per subscription and generate the explanations for all the transactions.

[Learn more](#)

Global explanation

☐ Off

Local explanation method

- ☐ SHAP (Kernel Explainer)
- ☒ LIME (enhanced)

Cancel Next

Set number of perturbations per record, it is a measure used to indicate how frequently each data record has been subject to modifications during the perturbation-based analysis.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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[Learn more](#)

LIME settings

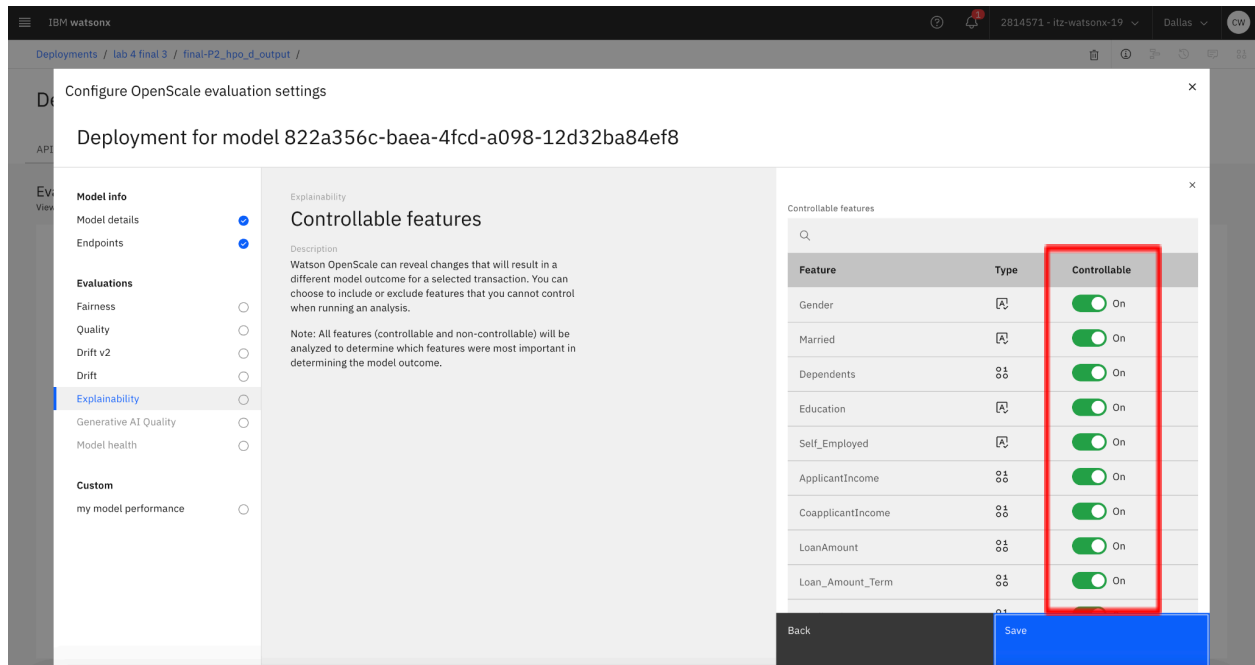
Number of perturbations per record

1000

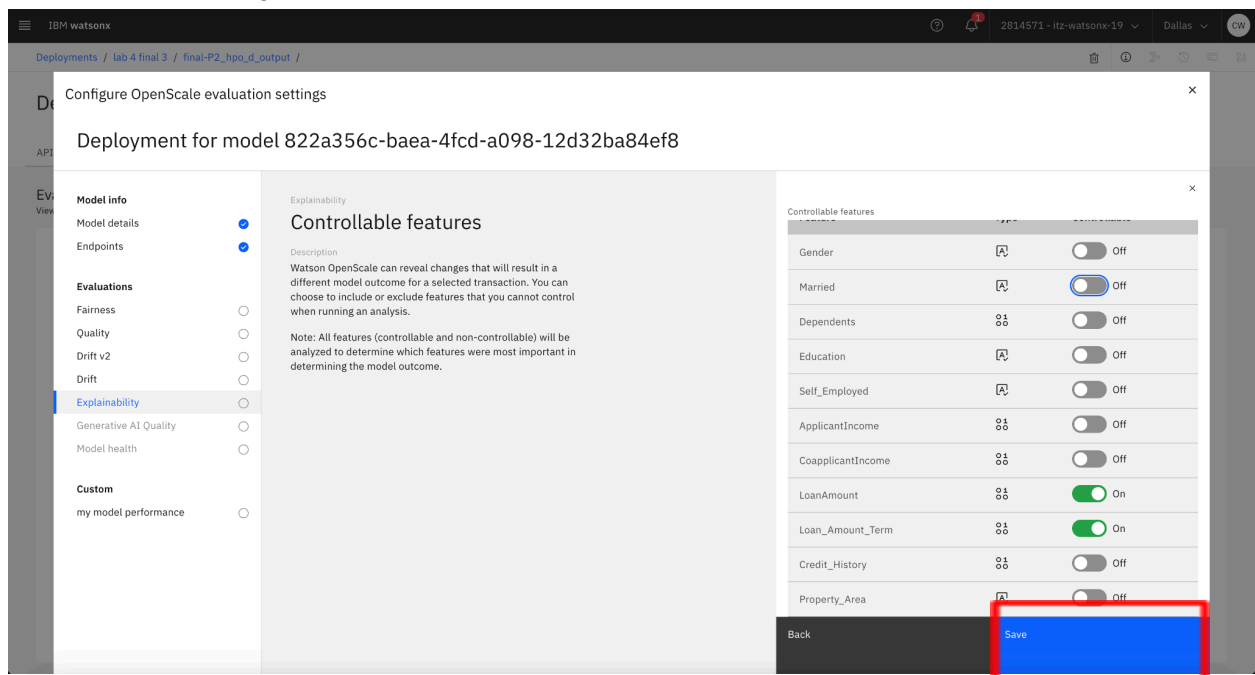
Set this to 0 to automatically choose a suitable count

Back Next

Select the features that are controllable.

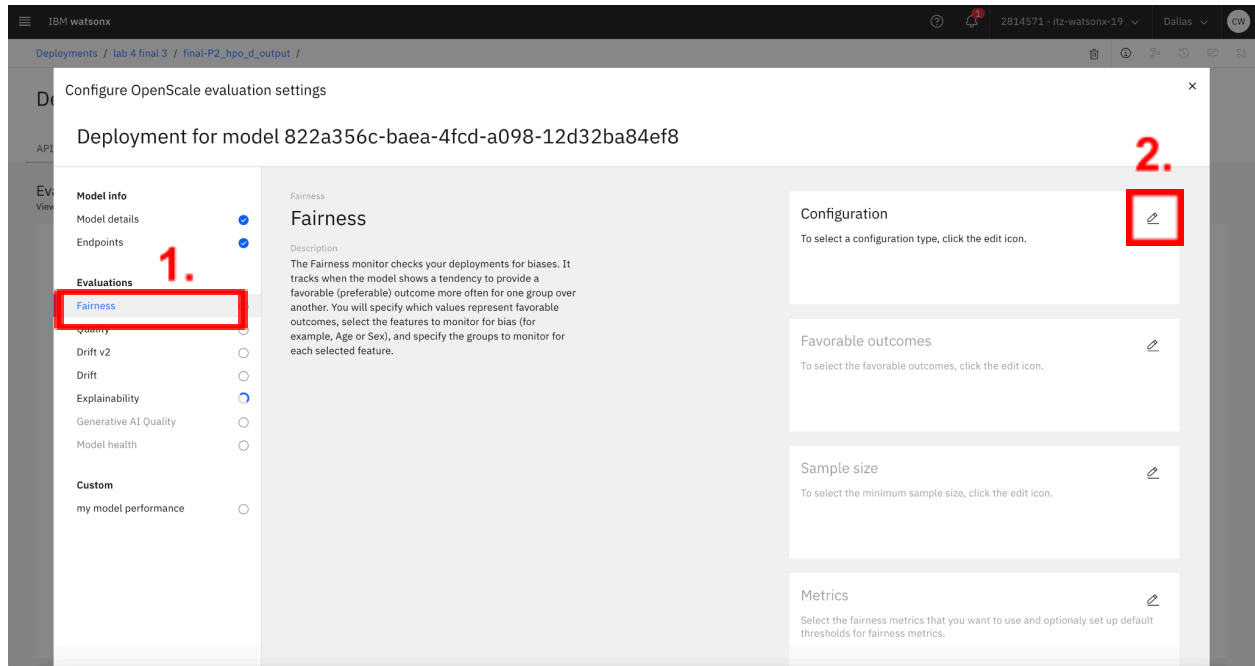


Once done choosing click save

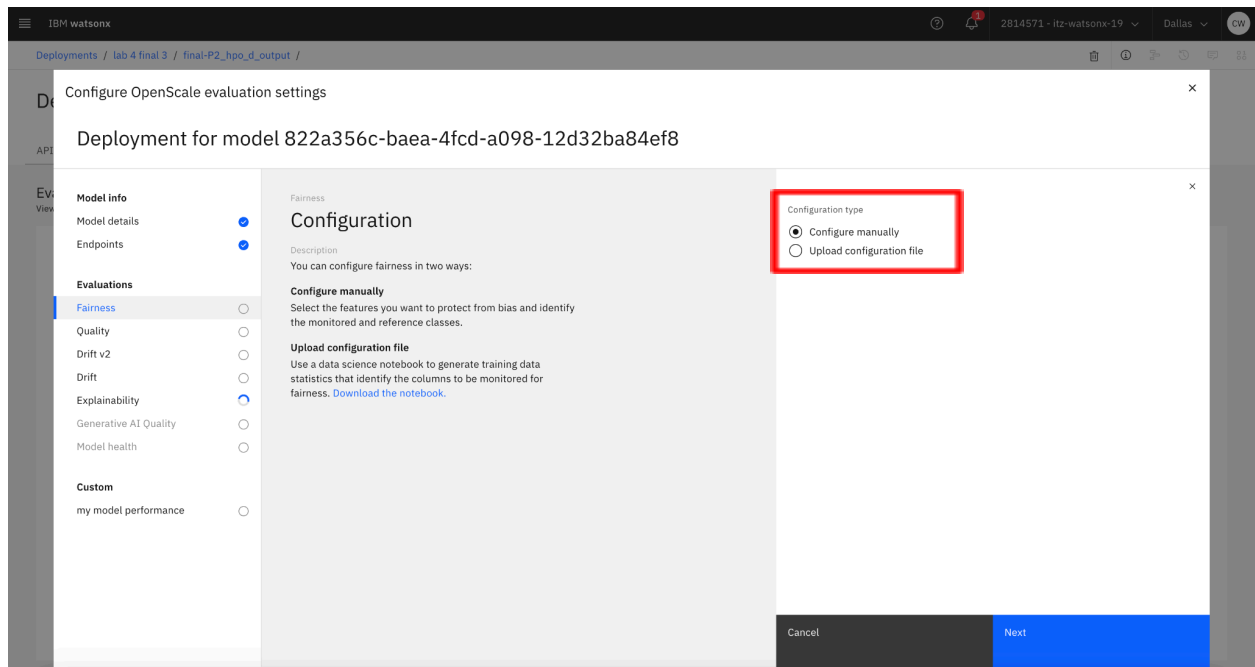


Part 3 : Configuring model fairness monitor

Go to Fairness under the Evaluations topic and you should be able to see this page. After that click on the edit icon of the configuration to start configuring the model fairness monitor.



Then choose configure manually under configuration type and click next to continue onto the next page.



Once you land on this page, choose the value of the outcomes according to the dataset ('Y' is favorable and 'N' in unfavorable)

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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Fairness

Select the favorable outcomes

Description
For each group, Watson OpenScale will calculate the percentage of transactions that receive a favorable outcome.

Select the values that represent favorable (preferable) outcomes. You can also add a value manually if it is not included in the list.

Add custom value
Enter a value

Favorable values 1 Unfavorable values 1

Values	Favorable	Unfavorable
N	☐	☒
Y	☒	☐

Once you are done selecting, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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Description
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Select the values that represent favorable (preferable) outcomes. You can also add a value manually if it is not included in the list.

Add custom value
Enter a value

Favorable values 1 Unfavorable values 1

Values	Favorable	Unfavorable
N	☐	☒
Y	☒	☐

Set the minimum sample size to be whatever number which is smaller than the prepared dataset. Once done, click next to continue onto the next page.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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Fairness

Sample size

Description
Ensure that your minimum sample size is large enough to accurately represent the variety of requests the deployment receives.

☐ Evaluate using balanced data set

Minimum sample size
100

Maximum sample size (optional)
10000

Back Next

Select the metrics which you want to monitor which in this case we only select Disparate impact. Once done, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
- Endpoints

Evaluations

- Fairness**
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Custom

- my model performance

Fairness

Metrics

Description
The Fairness monitor tracks multiple fairness metrics. You can select the metrics that will be monitored across the features in the future, and set up some default thresholds. You can always overwrite the default thresholds when setting up the monitored features. Disparate impact is pre-selected as the default fairness metric.

Metric thresholds track when a metric value is outside of an acceptable range. There are two types of thresholds.

Lower threshold - High metric value is better
Upper threshold - Low metric value is better

At least one metric must have a threshold. Metrics without a threshold will not trigger alerts.

Select monitored metrics

Metrics generated with payload data, training data, and balanced data ⓘ

- ☒ Disparate impact
- ☐ Statistical parity difference
- ☐ Impact score

Metrics generated with feedback data ⓘ

- ☐ Average odds difference
- ☐ Average absolute odds difference
- ☐ False negative rate difference
- ☐ False positive rate difference
- ☐ False discovery rate difference
- ☐ False omission rate difference
- ☐ Error rate difference

Back Next

Set the threshold for the metrics we selected previously which in this place is Disparate impact. Once done, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

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Fairness

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Metric thresholds track when a metric value is outside of an acceptable range. There are two types of thresholds.

Lower threshold - High metric value is better
Upper threshold - Low metric value is better

At least one metric must have a threshold. Metrics without a threshold will not trigger alerts.

Disparate impact

Lower threshold for fairness value

80

Upper threshold for fairness value

120

Other metrics

(Difference metrics are not selected.)

Back Next

Select field for tendency monitoring to make sure that model provides output which has acceptable fairness level. Once done, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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Custom

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Fairness

Select the fields to monitor

Description

For each field you select, Watson OpenScale will monitor the deployed model's tendency to provide a favorable (preferred) outcome for one group over another.

If you select a field that is not a feature column, called a added field, Watson OpenScale will look for indirect bias by finding associated values in the feature columns. For example, the profession "student" may imply a younger individual even though the Age field was excluded from model training.

Recommended features

Watson OpenScale analyzed your training data to recommend which features should be monitored for fairness. These features are identified in the Recommended column.

Select one or more fields

Selected fields 1

Fields	Recommended	Type
☒ Gender	☒	
☐ Married	☐	
☐ Dependents	☐	
☐ Education	☐	
☐ Self_Employed	☐	
☐ ApplicantIncome	☐	
☐ CoapplicantIncome	☐	
☐ LoanAmount	☐	
☐ Loan_Amount_Term	☐	

Items per page: 25 1 - 12 of 12 items 1 x of 1 page

Back Next

Select value from the selected field to be monitored or to be a reference where the output value from those two groups will be compared in order to check for potential bias. Once done, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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Fairness

Specify the monitored groups for [Gender]

Description: Select the groups to monitor.

The percentage of favorable outcomes delivered to the monitored groups will be compared to the percentage of favorable outcomes delivered to the remaining groups (the reference groups) to check for potential bias.

A fairness score of 100% implies that the monitored group and reference group received an equal number of favorable values. Likewise, a fairness score of 50% implies that the monitored group received half as many favorable outcomes as the reference group.

Set the fairness alert threshold to track when the fairness value falls below an acceptable level.

Recommended groups

Watson OpenScale analyzed this feature to provide recommendations for group assignments. Less frequently occurring values in the training data may be suggested as monitored groups. Always review these recommended selections before proceeding.

Select the groups to monitor [Gender]

Add custom value

Enter a value

Add value

Monitored group 1 Reference group 1

Values	Monitored	Reference	Recommended
Female	☒	☐	Monitored
Male	☐	☒	Reference

Back Next

Select threshold for the monitored field which in this place we will use the default threshold. Once done, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
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Evaluations

- Fairness
- Quality
- Drift v2
- Drift
- Explainability
- Generative AI Quality
- Model health

Custom

- my model performance

Fairness

Specify the monitored groups for [Gender]

Description: Select the groups to monitor.

The percentage of favorable outcomes delivered to the monitored groups will be compared to the percentage of favorable outcomes delivered to the remaining groups (the reference groups) to check for potential bias.

A fairness score of 100% implies that the monitored group and reference group received an equal number of favorable values. Likewise, a fairness score of 50% implies that the monitored group received half as many favorable outcomes as the reference group.

Set the fairness alert threshold to track when the fairness value falls below an acceptable level.

Recommended groups

Watson OpenScale analyzed this feature to provide recommendations for group assignments. Less frequently occurring values in the training data may be suggested as monitored groups. Always review these recommended selections before proceeding.

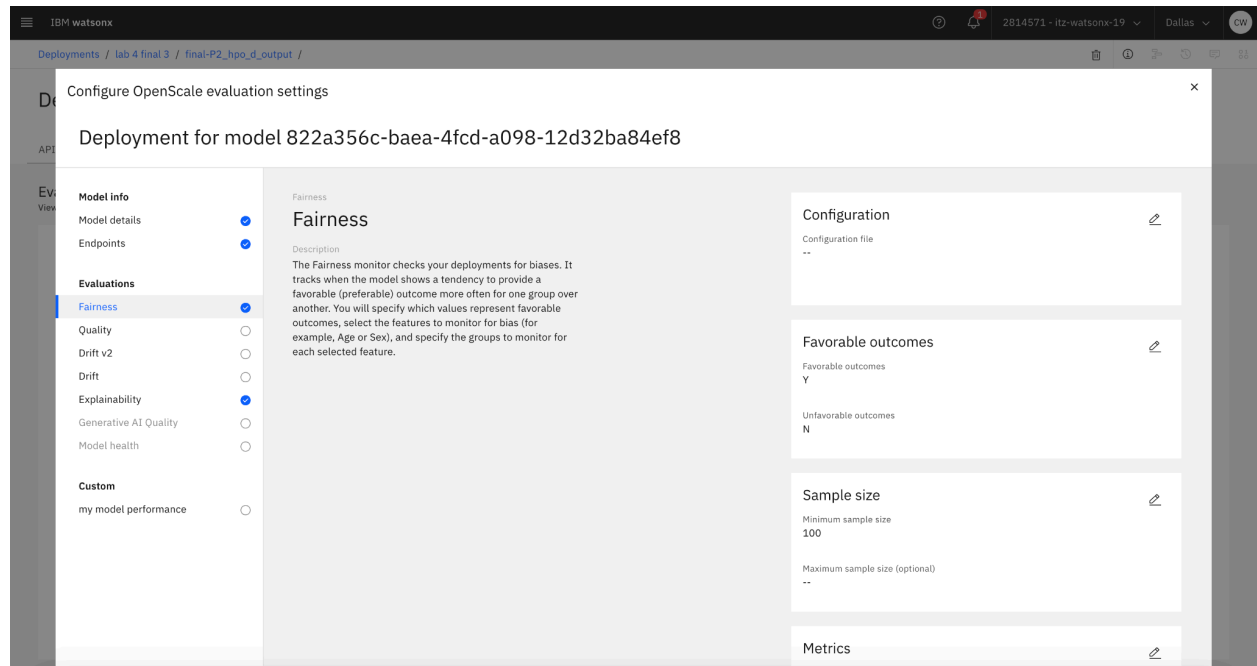
Set up threshold for feature [Gender]

- ☒ Use default thresholds
- ☐ Set up different thresholds

Disparate impact 80 - default

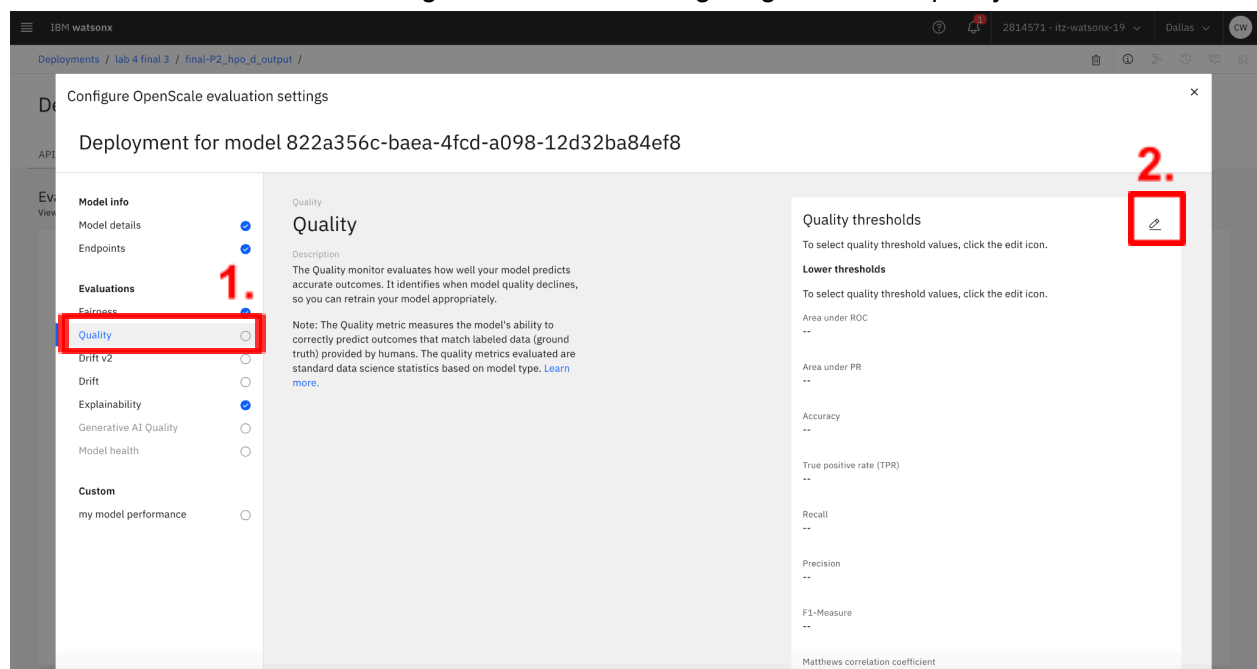
Back Save

Now you have finished your model fairness monitor setup and should be able to see this page.



Part 4 : Configuring model quality monitor

Go to Quality under the Evaluations topic and you should be able to see this page. After that click on the edit icon of the configuration to start configuring the model quality monitor.



Once you click on the configuration icon, you should enter this page. In this page you can adjust the thresholds for model quality as you prefer but for now, we will leave it as it is. Click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
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Custom

- my model performance

Quality

Quality thresholds

Description

Area under ROC

Quality metrics are based on model type. Metric thresholds track when a metric value is outside of an acceptable range. There are two types of thresholds.

- Lower threshold - High metric value is better
- Upper threshold - Low metric value is better

At least one metric must have a threshold. Metrics without a threshold will not trigger alerts.

Lower thresholds

Area under ROC ①

0.8

Area under PR ①

0.8

Accuracy ①

0.8

True positive rate (TPR) ①

0.8

Recall ①

0.8

Precision ①

0.8

F1-Measure ①

0.8

Cancel Next

Set the minimum sample size to be whatever number which is smaller than the prepared dataset. Once done, click next to continue onto the next page.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
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Custom

- my model performance

Quality

Sample size

Description

Ensure that your minimum sample size is large enough to accurately represent the variety of requests the deployment receives.

Minimum sample size

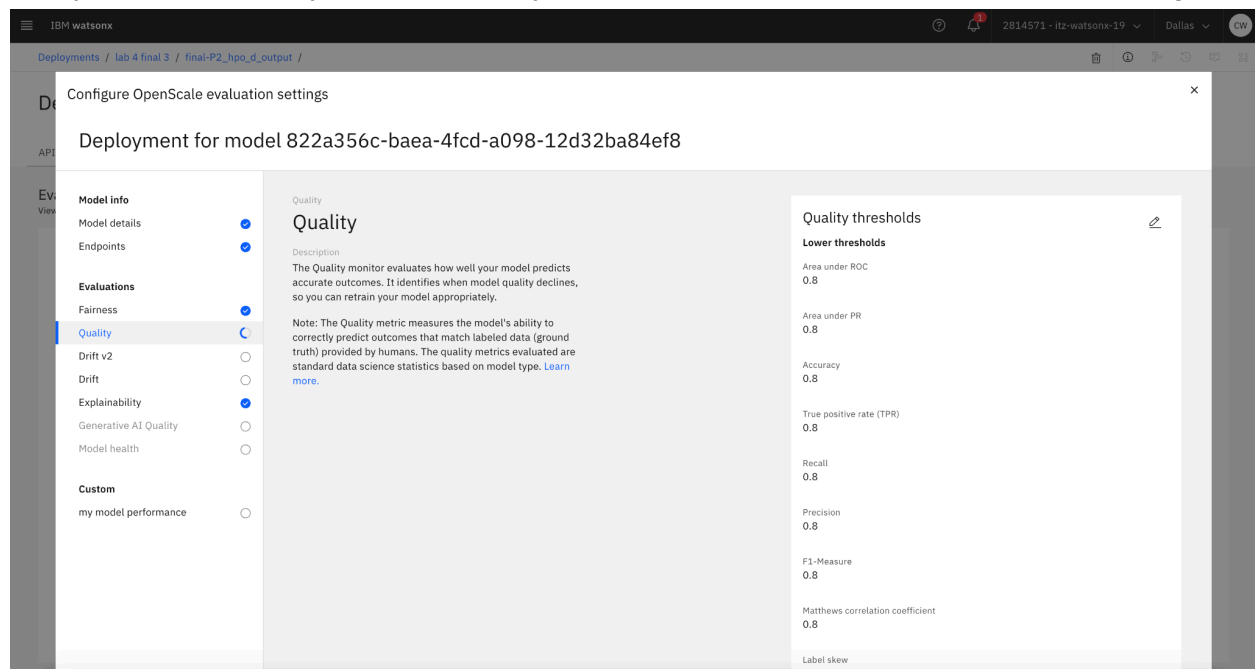
100

Maximum sample size (optional)

1000

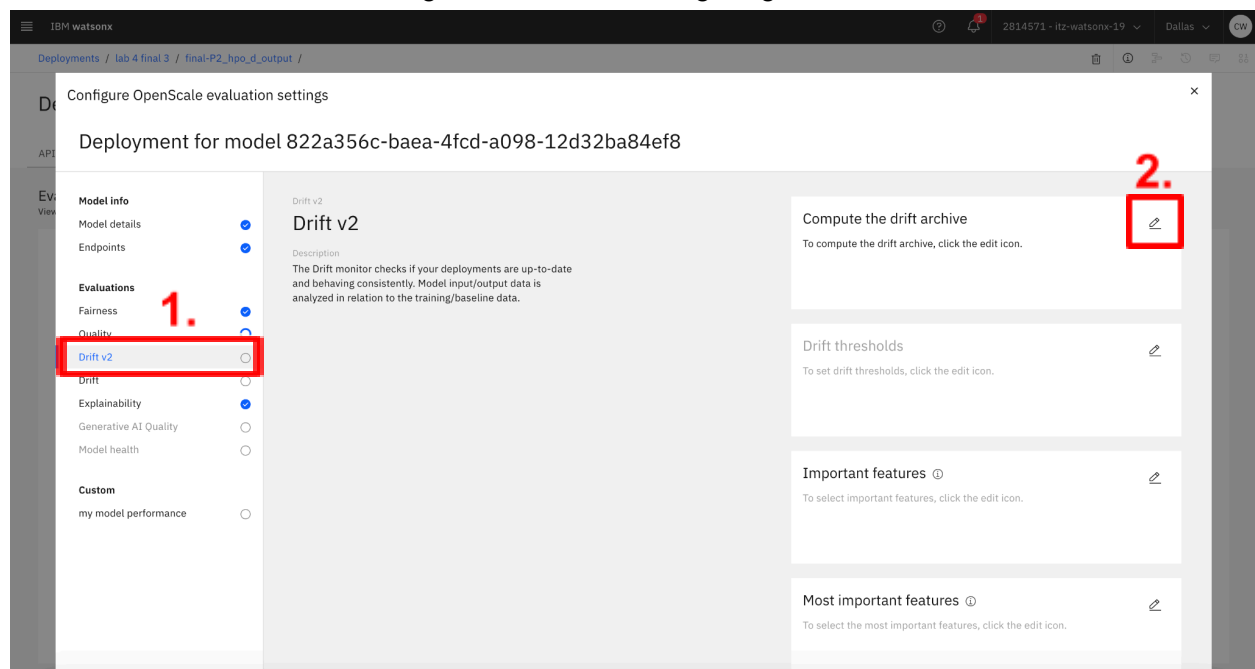
Back Save

Now you have finished your model quality monitor setup and should be able to see this page.



Part 5 : Configuring model Drift v2 monitor

Go to Drift v2 under the Evaluations topic and you should be able to see this page. After that click on the edit icon of the configuration to start configuring the model drift v2 monitor.



Select the compute option to be within Watson OpenScale. Once done, click next to continue.

IBM watsonx

Deployments / lab 4 final 3 / final-P2_hpo_d_output /

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

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Custom

my model performance

Compute the drift archive

Description

Watson OpenScale analyzes your training data to determine the data distributions of the features. You can perform this analysis in Watson OpenScale or you can do it yourself using a custom notebook.

Compute in Watson OpenScale

If you connected your training data to Watson OpenScale, and it is less than 500MB, use this option.

Compute in a data science notebook

If you did not connect your training data to Watson OpenScale or it exceeds 500MB, use this option. [Download the notebook.](#)

Compute option

☒ Compute in Watson OpenScale

☐ Compute in a data science notebook

Cancel Next

Drift upper thresholds track the degree of change of model input and output from its training data. Set it as you prefer but in this case we will leave it as it is. Click next to continue.

IBM watsonx

Deployments / lab 4 final 3 / final-P2_hpo_d_output /

Configure OpenScale evaluation settings

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Custom

my model performance

Drift thresholds

Description

Drift metrics track the degree of change in model input and output in relation to the training or baseline data. Classification model metrics include:

Output drift

Measures the change in the model confidence distribution. [Read a sample scenario.](#)

Model quality drift

Measures the drop in accuracy by comparing the estimated runtime accuracy to the training accuracy. Ground truth labels are not required. [Read a sample scenario.](#)

Feature drift

Measures the change in value distribution for important features. [Read a sample scenario.](#)

Prediction drift

Measures the change in model prediction classes distribution. [Read a sample scenario.](#)

Upper thresholds

Output drift ⓘ

0.05

Model quality drift ⓘ

0.05

Feature drift ⓘ

0.05

Prediction drift ⓘ

0.05

Back Next

Select the important features which impact the model from the list.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

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Custom

- my model performance

Important features

Description

The system uses feature importance to determine the impact of feature drift on the model. For example, a small amount of drift in an important feature may have a bigger impact on the model than a moderate amount of drift in a less important feature.

Note: When SHAP is configured, the important features are automatically detected using the model's global explanation. As SHAP is not configured, you must indicate the important features manually.

Select from list

Select up to 100 important features.

Upload list

Generate a global explanation and upload the list of important features as a json file. Some snippets/examples to extract this information for popular ML frameworks has been provided on this [wiki](#).

Selection options

- Select from list
- Upload list

Select important features

Filter table

Features (12)	Type
☐ ApplicantIncome	📊
☐ CoapplicantIncome	📊
☐ Credit_History	📊
☐ Dependents	📊
☐ Education	📄
☐ Gender	📄
☐ LoanAmount	📊
☐ LoanAmount_Term	📊

Back Next

Once done click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

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Evaluations

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Custom

- my model performance

Important features

Description

The system uses feature importance to determine the impact of feature drift on the model. For example, a small amount of drift in an important feature may have a bigger impact on the model than a moderate amount of drift in a less important feature.

Note: When SHAP is configured, the important features are automatically detected using the model's global explanation. As SHAP is not configured, you must indicate the important features manually.

Select from list

Select up to 100 important features.

Upload list

Generate a global explanation and upload the list of important features as a json file. Some snippets/examples to extract this information for popular ML frameworks has been provided on this [wiki](#).

Selection options

- Select from list
- Upload list

Select important features

Filter table

Features (12)	Type
☒ ApplicantIncome	📊
☐ CoapplicantIncome	📊
☒ Credit_History	📊
☐ Dependents	📊
☐ Education	📄
☐ Gender	📄
☒ LoanAmount	📊
☒ LoanAmount_Term	📊

Back Next

Select the most important features which have a high impact on the model from the list.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

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- Custom**
- my model performance

Drift v2

Most important features

Description

Select up to 10 most important features to perform root cause analysis on the drift issues detected. The system will generate insights based on this selection. [Read a sample scenario.](#)

Select most important features

Filter table

Features (4)	Type
☒ ApplicantIncome	01
☒ Credit_History	01
☒ LoanAmount	01
☒ LoanAmount_Term	01

Back Next

Set the minimum sample size to be whatever number which is smaller than the prepared dataset. Once done, click save to continue onto the next page.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
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Drift v2

Sample size

Description

The minimum sample size is the lowest number of transactions that will be evaluated during drift evaluation. If the number of new records received is below the sample size, additional records from previous evaluations will be added to meet the required sample size. Maximum sample size is the maximum number of transactions that will be evaluated during drift analysis. If maximum sample size is not provided, all the records added since the last drift evaluation will be analyzed to compute drift.

Minimum sample size

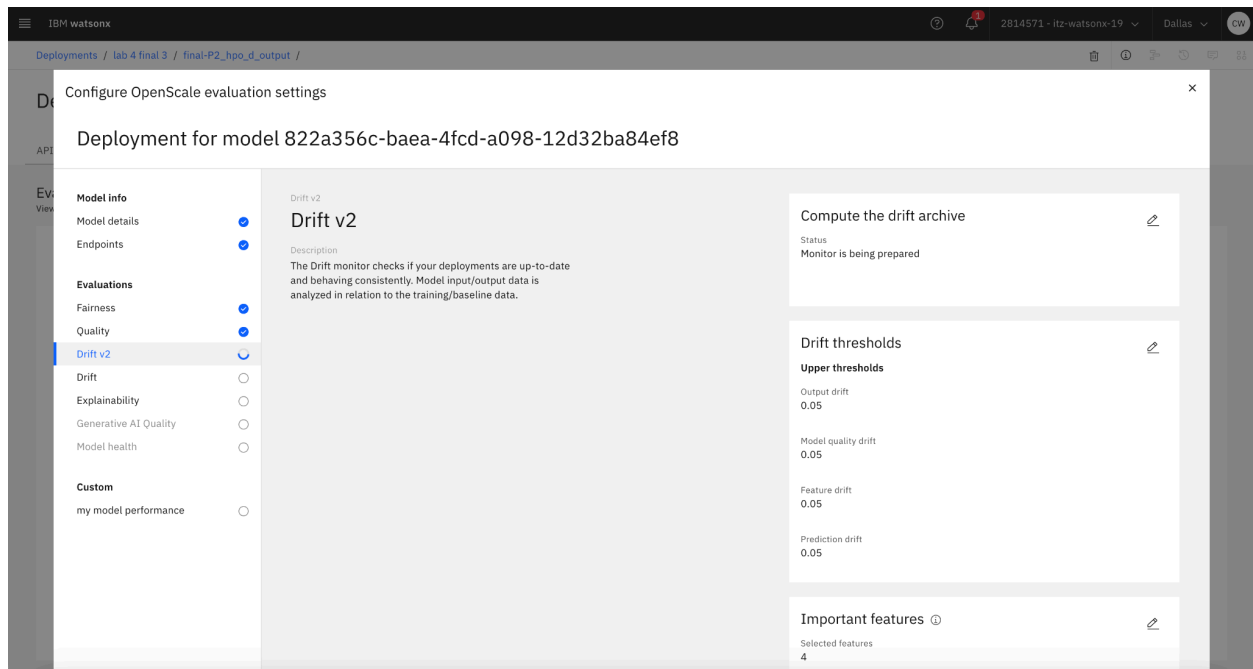
100

Maximum sample size (optional)

10000

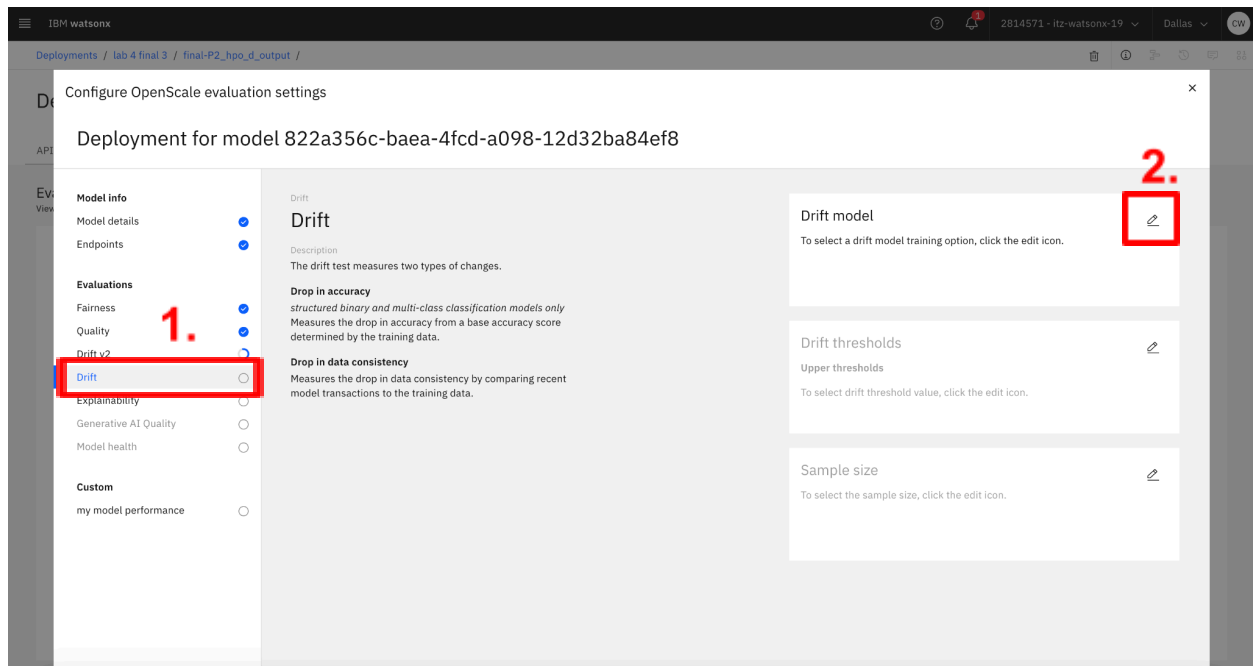
Back Save

Now you have finished your model drift v2 monitor setup and should be able to see this page.



Part 6 : Configuring model Drift monitor

Go to Drift under the Evaluations topic and you should be able to see this page. After that click on the edit icon of the configuration to start configuring the model drift monitor.



Select the training option to be within Watson OpenScale. Once done, click next to continue.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
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Custom

- my model performance

Drift

Train a drift model

Description

Watson OpenScale can analyze your training data to build a drift detection model and establish an accuracy and data consistency baseline.

You can train the model in Watson OpenScale or you can do it yourself using a custom notebook.

Train in Watson OpenScale

Use this option when:

- The deployment is hosted by Watson Machine Learning
- You provided a connection to the training data
- The training data is less than 500 MB

Train in a data science notebook

Run a [custom notebook](#) to generate the drift detection model.

Training option

- ☒ Train in Watson OpenScale
- ☐ Train in a data science notebook

Cancel Next

Setup drift upper thresholds which “drop in accuracy” means to measure the drop in accuracy from a base accuracy score determined by the training data and “drop in data consistency” means to measure the drop in data consistency by comparing recent model transactions to the training data.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
- Endpoints

Evaluations

- Fairness
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- Drift
- Explainability
- Generative AI Quality
- Model health

Custom

- my model performance

Drift

Drift thresholds

Description

Drift metrics track the degree of change in accuracy and data consistency based on the accuracy and data consistency at training time. There is one type of threshold:

- Upper threshold - Low metric value is better

The drift calculation is based on the environment type.

Pre-production models

Upload labeled test data to determine the accuracy score for the model. This score is compared to the baseline accuracy score established by the model's performance on the training data.

Production models

Watson OpenScale will automatically label incoming model transactions as correct or incorrect using a custom drift model. This labeled data is used to determine an estimated accuracy.

Upper thresholds

Drop in accuracy ⓘ

10 %

Drop in data consistency ⓘ

10 %

Back Next

Set the minimum sample size to be whatever number which is smaller than the prepared dataset. Once done, click save to continue onto the next page.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

- Model details
- Endpoints

Evaluations

- Fairness
- Quality
- Drift v2
- Drift**
- Explainability
- Generative AI Quality
- Model health

Custom

- my model performance

Sample size

Description

Ensure that your sample size is large enough to be representative of the requests the deployment receives.

If the number of new records received exceeds the sample size, the additional records will also be evaluated. If the number of new records received is below the sample size, additional records from previous evaluations will be added to meet the required sample size.

Minimum sample size

100

Maximum sample size (optional)

10000

Back Save

Now you have finished your model drift monitor setup and should be able to see this page.

Configure OpenScale evaluation settings

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8

Model info

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Evaluations

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- Explainability
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- Model health

Custom

- my model performance

Drift

Description

The drift test measures two types of changes.

Drop in accuracy

structured binary and multi-class classification models only

Measures the drop in accuracy from a base accuracy score determined by the training data.

Drop in data consistency

Measures the drop in data consistency by comparing recent model transactions to the training data.

Drift model

Status

Monitor is being prepared

Drift thresholds

Upper thresholds

Drop in accuracy

10%

Drop in data consistency

10%

Sample size

Minimum sample size

100

Maximum sample size (optional)

--

Back Save

Part 7 : Start evaluation

Once you are done configuring the evaluation monitor you should be on this page. Click on evaluate now under actions dropdown.

The screenshot shows the IBM watsonx interface. The top navigation bar includes the IBM watsonx logo, a search icon, a notification bell, and user information (2814571 - itz-watsonx-19, Dallas, CW). The breadcrumb trail is 'Deployments / lab 4 final 3 / final-P2_hpo_d_output /'. The main heading is 'Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8', with status indicators 'Deployed' and 'Online'. Below this is a tabbed interface with 'API reference', 'Test', 'Evaluations' (selected), 'Transactions', and 'AI Factsheet'. The 'Evaluations' tab displays a summary of the last evaluation, which is '--'. It shows 'Test data set' as '--' and 'Number of explanations' as '0'. A circular gauge indicates '0 Tests run'. To the right, 'Tests passed' and 'Tests failed' are both '0'. An 'Actions' dropdown menu is visible in the top right corner. A message at the bottom states: 'Awaiting test data to evaluate. Select the Actions button to evaluate now.'

Once you click evaluate now, click import from CSV file to import the test data using .csv file.

This screenshot shows the same IBM watsonx interface as the previous one, but with the 'Import test data' dialog box open. The dialog has a title 'Import test data' and a description explaining that watsonx.governance evaluates production models for fairness and drift using logged scoring requests. It provides instructions on how to import test data by uploading a CSV file or connecting to a database table. The 'Import' section of the dialog is highlighted with a red box, showing a 'Choose an option' dropdown menu with two visible options: 'from database or cloud storage' and 'from CSV file'. The 'Upload and evaluate' button is visible at the bottom right of the dialog.

upload the test data into the red area.

The screenshot shows the IBM Watsonx interface. The main panel displays the 'Evaluations' tab for a specific model deployment. It shows a 'Last evaluation: --' status, a 'Test data set' of '--', and '0' tests run. A modal window titled 'Import test data' is open on the right. The modal has a description of the evaluation process and an 'Import' section with a dropdown menu set to 'from CSV file'. Below the dropdown is a red rectangular area with the text 'Drag test data file here or browse'. At the bottom of the modal are 'Close' and 'Upload and evaluate' buttons.

Once you upload the data, your file name should appear on the test data set. Click upload and evaluate to evaluate.

This screenshot shows the same IBM Watsonx interface after the test data has been uploaded. The 'Test data set' field now displays the filename 'dataset_loan_example_with_loanID_Ylabel4 copy.csv'. The 'Import' dropdown in the modal is still set to 'from CSV file'. The 'Upload and evaluate' button at the bottom right of the modal is now highlighted in blue, indicating it is the next step to click.

Part 8 : Evaluation result

Once done with evaluate, you should be able to get the evaluation result as shown below. You can click on the arrow button inside each metrics monitor to find further information for that monitor type.

Deployment for model 822a356c-baea-4fcd-a098-12d32ba84ef8 Deployed Online

API reference | Test | **Evaluations** | Transactions | AI Factsheet

Last evaluation: Mon, Oct 7, 2024, 1:10 PM +07

Test data set
dataset_loan_example_with_loanID_Ylabel4 copy.csv

Number of explanations
2

Tests passed: 1
Tests failed: 3

4 Tests run

Fairness No alerts →

Total alerts: 0
Alerts for Gender: 0
Gender: Gender

Metric	Score	Violation
Disparate impact	100.00%	none

Quality Alerts triggered →

Alerts: 1

Metric	Score	Violation
True positive rate (TPR)	0.98	none
Area under ROC	0.94	none

Drift v2 →

Drift v2 evaluation failed.
Drift computation failed. Argument 'value' has incorrect type (expected int, got numpy.int64)

Associated group: Female
84 records evaluated

Area under PR	0.95	none
Recall	0.98	none
Brier score	0.21	none

566 records evaluated

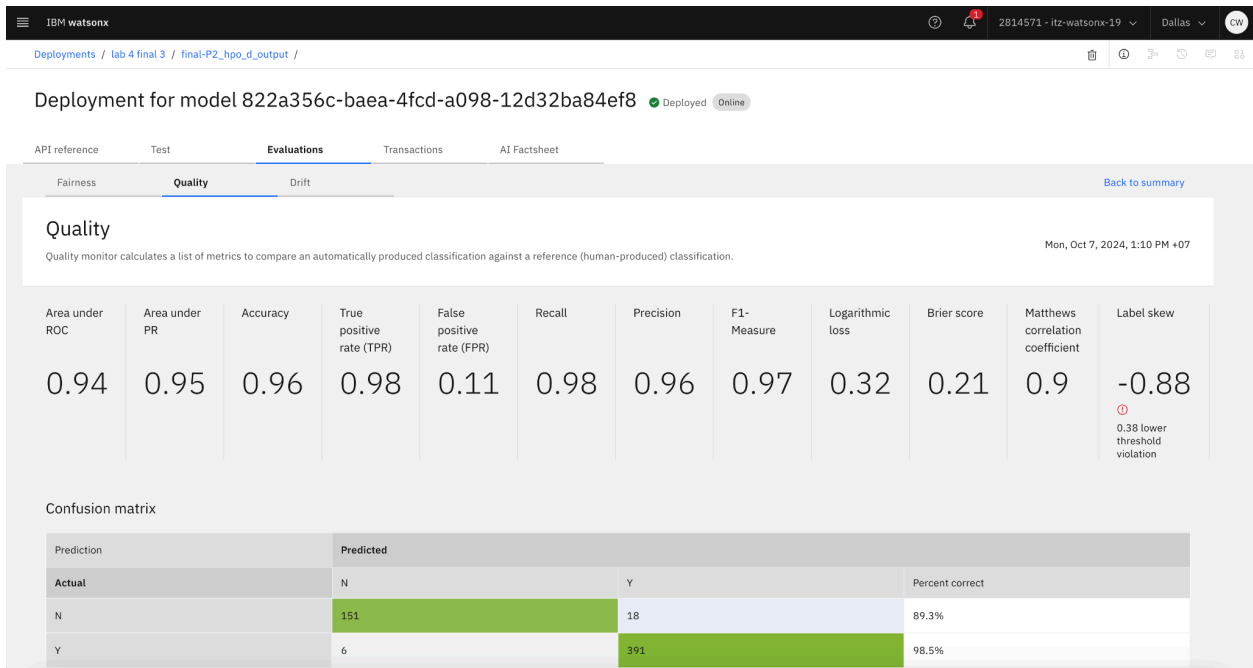
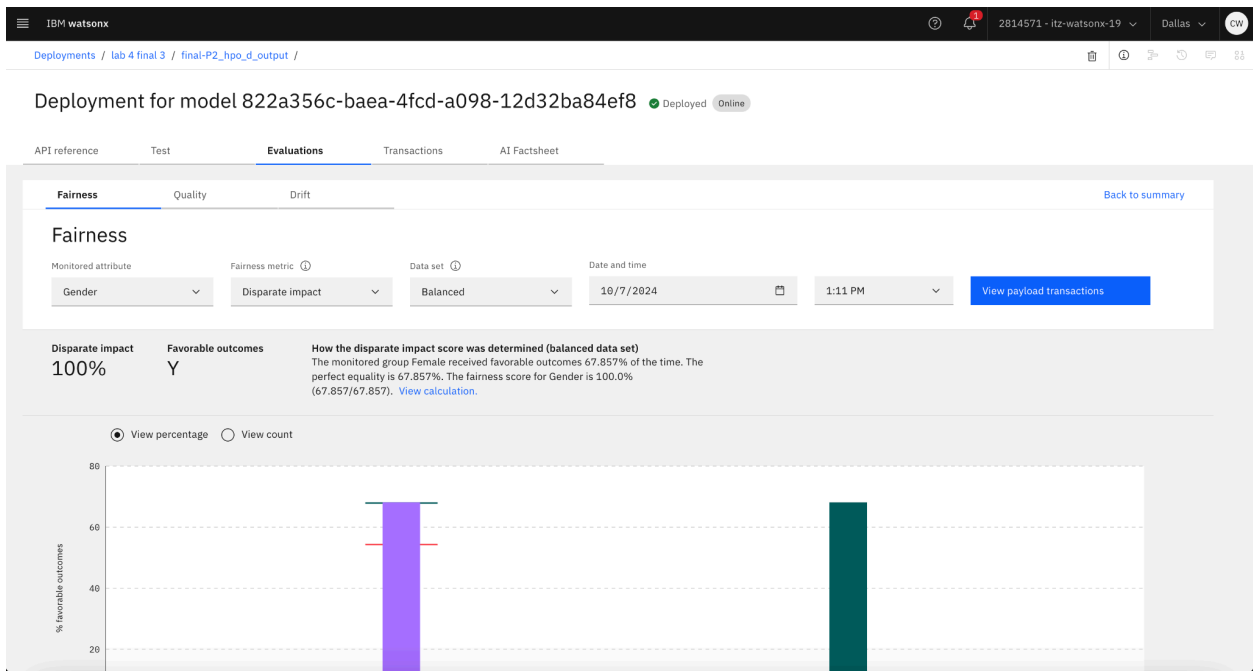
Drift Alerts triggered →

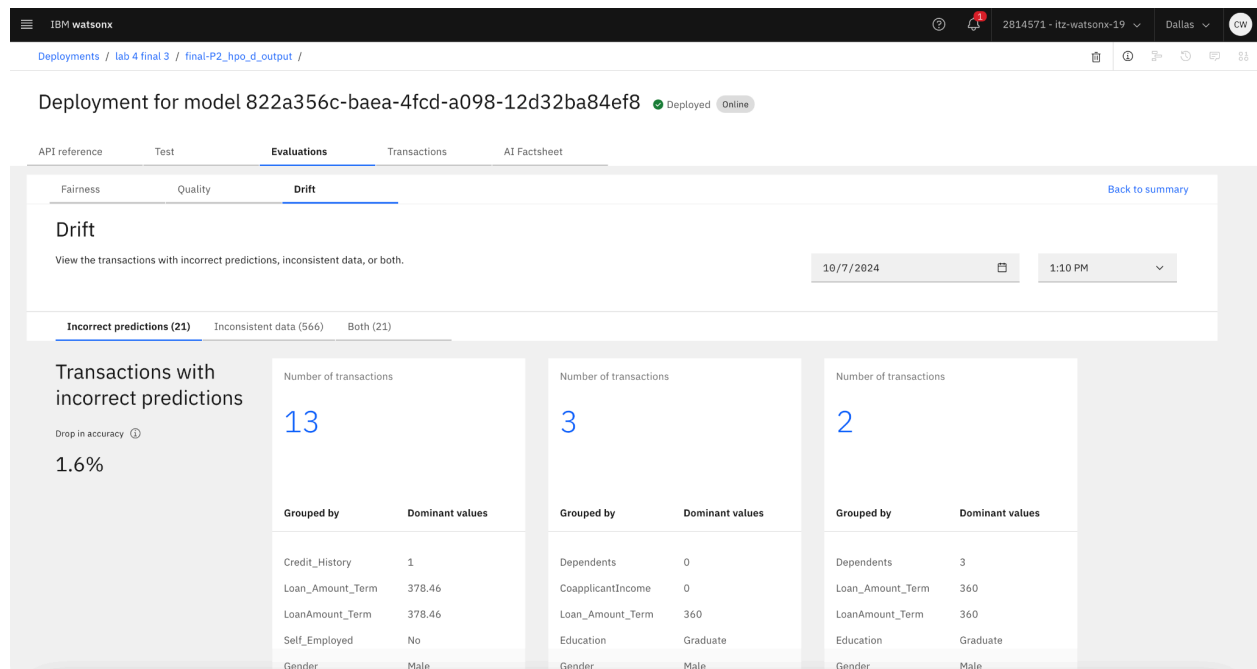
Alerts: 1

Metric	Score	Violation
Drop in data consistency	100.00%	90.00%
Drop in accuracy	1.63%	none
Evaluated accuracy	96.29%	--
Base accuracy	97.92%	--

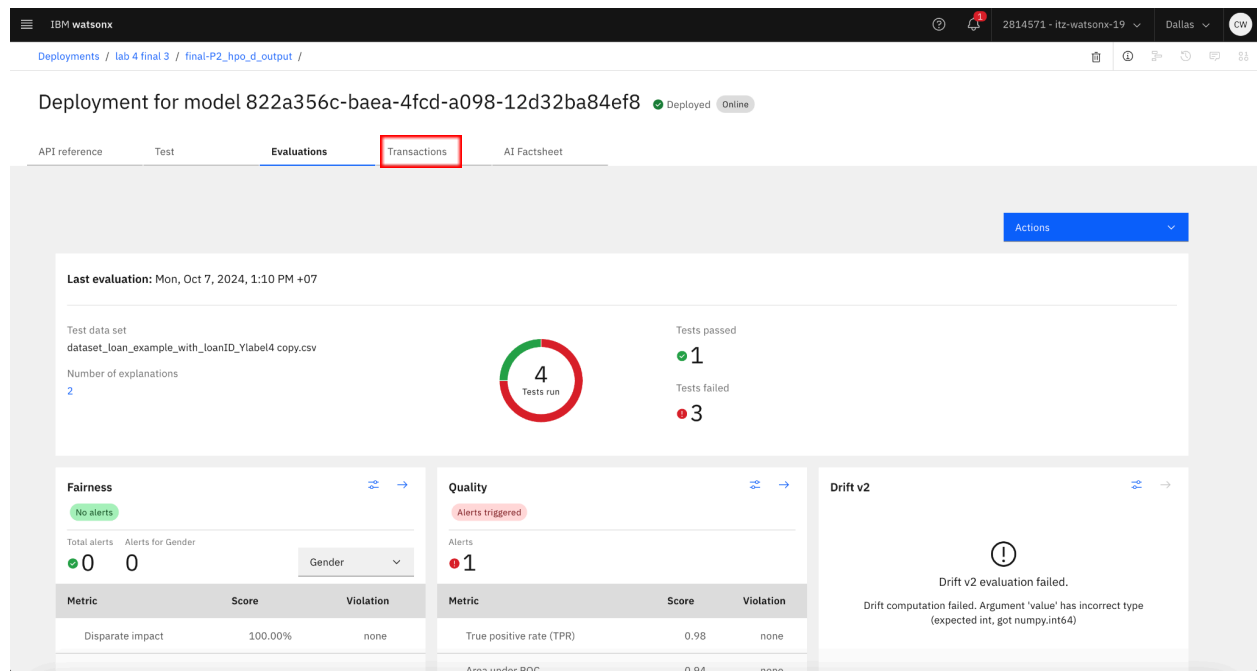
566 records evaluated

Here is the example of further information for each evaluation metrics.

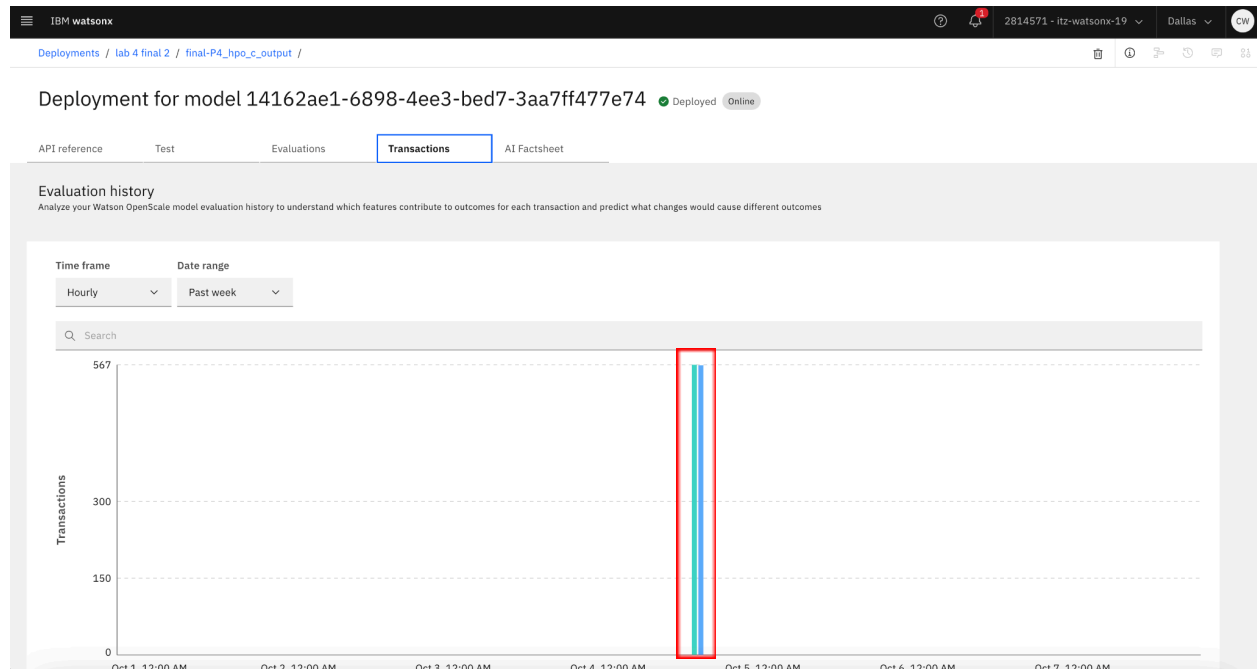




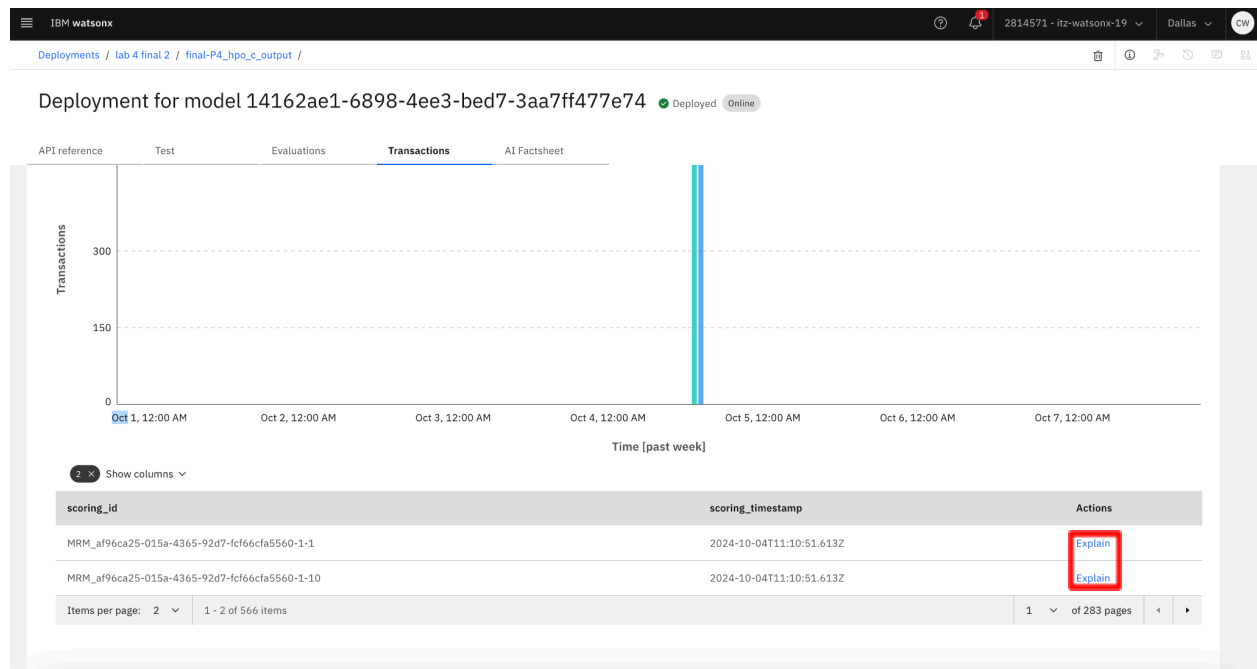
For model explainability, click on transactions tab in the red area



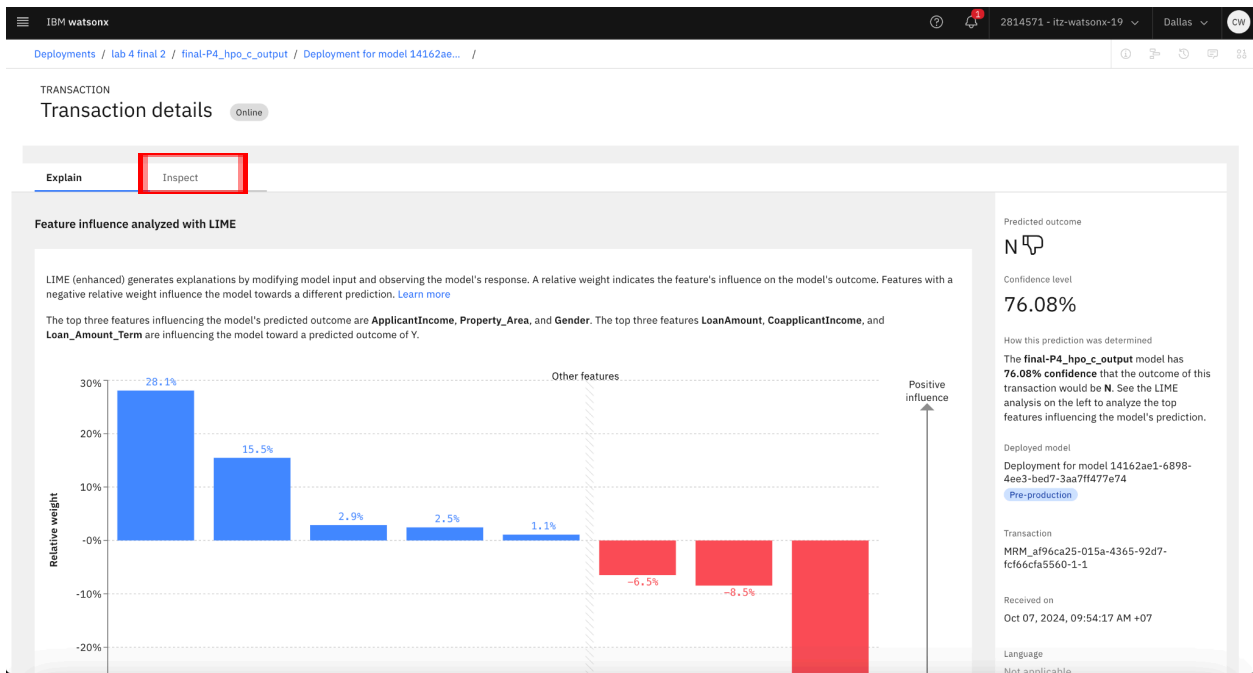
Once you click on transactions you should see this page. Then, click on the bar chart to view its detail



After you click on the bar chart the list underneath should appear. Click explain to view model explainability.



Now you are able to observe the model explainability through feature importance. If you want to see how the change of the feature affects the model you can use the 'inspect' tab to observe.



Once you click inspect, you should be entering this page. In this page you can config the feature values to predict a different outcome by adjusting the feature value in the red area.

Explain Inspect

Reaching a different predicted outcome

For the model to have predicted a different outcome for this transaction, the value of all listed features would need to change to the indicated minimum value. Note that changing a feature value by more than the minimum value may affect the minimum change of other features for the model to predict a different outcome. Higher feature importance numbers indicate a greater likelihood of changing the prediction.

☒ Analyze controllable features only ⓘ

Feature	Original value	New value
Gender	Male	Male
Married	Yes	Yes
Dependents	1	1 ⓘ
Education	Graduate	Graduate
Self_Employed	No	No
ApplicantIncome	4583	4583 ⓘ
CoapplicantIncome	1508	1508 ⓘ
LoanAmount	128	128 ⓘ
Predicted outcome	N	N
Confidence	79.34%	79.34%

Value for a different outcome ⓘ Importance

Want to know what feature values will result in a different outcome? Analyze the model to get a suggestion. This analysis will take a few minutes.

[Run analysis](#)

Once you are done configuring the value, click “Score new values” and then click “Run analysis”.

IBM watsonx

2814571 - itz-watsonx-19DallasCW

Deployments / lab 4 final 3 / final-P2_hpo_d_output / Deployment for model 94b076c... /

ExplainInspect

Reaching a different predicted outcome

For the model to have predicted a different outcome for this transaction, the value of all listed features would need to change to the indicated minimum value. Note that changing a feature value by more than the minimum value may affect the minimum change of other features for the model to predict a different outcome. Higher feature importance numbers indicate a greater likelihood of changing the prediction.

☒ Analyze controllable features only ⓘ

Feature	Original value	New value	Value for a different outcome ⓘ	Importance
Gender	Male	Male		
Married	Yes	Yes		
Dependents	1	1 ⓘ		
Education	Graduate	Graduate		
Self_Employed	No	Yes ⓘ		
ApplicantIncome	4583	4583 ⓘ		
CoapplicantIncome	1508	1508 ⓘ		
LoanAmount	128	128 ⓘ		
Predicted outcome	N			
Confidence	79.34%			

Score new values

Reset

Want to know what feature values will result in a different outcome? Analyze the model to get a suggestion. This analysis will take a few minutes.

Run analysis

After you click run analysis, you should be able to see the outcome in the red area.

IBM watsonx

2814571 - itz-watsonx-19DallasCW

Deployments / lab 4 final 3 / final-P2_hpo_d_output / Deployment for model 94b076c... /

ExplainInspect

Reaching a different predicted outcome

For the model to have predicted a different outcome for this transaction, the value of all listed features would need to change to the indicated minimum value. Note that changing a feature value by more than the minimum value may affect the minimum change of other features for the model to predict a different outcome. Higher feature importance numbers indicate a greater likelihood of changing the prediction.

☒ Analyze controllable features only ⓘ

Feature	Original value	New value	Value for a different outcome ⓘ	Importance
ApplicantIncome	4583	4583 ⓘ	4666.5	1.00
Gender	Male	Male	Male	0.00
Married	Yes	Yes	Yes	0.00
Dependents	1	1 ⓘ	1	0.00
Education	Graduate	Graduate	Graduate	0.00
Self_Employed	No	No	No	0.00
CoapplicantIncome	1508	1508 ⓘ	1508	0.00
LoanAmount	128	128 ⓘ	128	0.00
Predicted outcome	N		Predicted outcome	
Confidence	79.34%		Confidence	

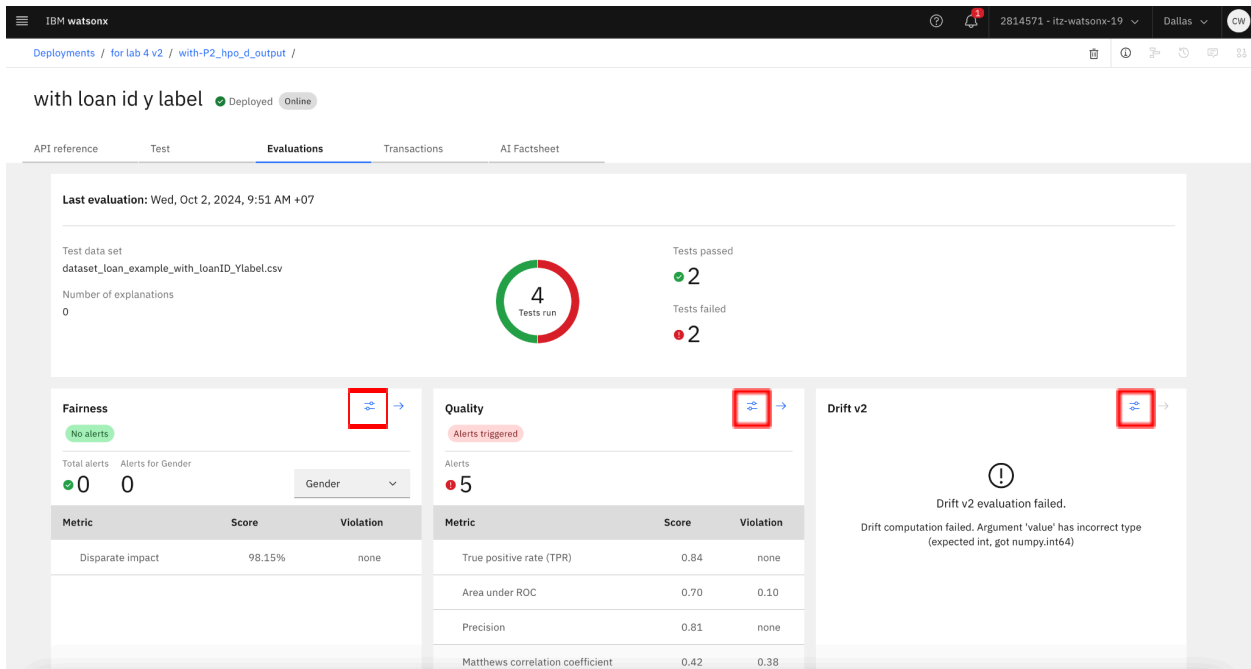
Predicted outcome

Confidence

Predicted outcome

Confidence

Furthermore, you can reconfigure the evaluation metrics monitor by clicking on the configure button in the red area.



Which once you clicked on the reconfigure button in the red area. Now you can reconfigure OpenScale evaluation settings.

